Free fall

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe and calculate ideal free fall near Earth.

SUCCESS CRITERIA - I can...

1. State the no-air-resistance assumption.
2. Use equal velocity changes in equal times.
3. Separate acceleration from mass and current velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “free fall” and “initial” to explain your model or result. Sentence starters are optional.

free fall: motion with gravity as the only force

initial: at the beginning of the interval; subscript i

acceleration: rate of change of velocity

01 Bring back an earlier idea

A 4 kg cart starts from rest under 24 N net force right for 3 s.
Find acceleration and final velocity.
State one assumption.

Word help: net force = the vector sum of all forces on one object

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_f = v_i + g\Delta t$.
Downward positive; g = 10 m/s²; gravity only.

Worked reasoning: A ball dropped from rest has downward speeds 0, 10, 20 m/s at 0, 1, 2 s in ideal free fall with g=10 m/s². Equal times give equal velocity changes.

Your turn: Sketch its velocity-time graph with downward positive.
Explain why the positions are not equally spaced.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

03 Look and explain

Two balls of different mass are released together in a vacuum from the same height.
Compare acceleration and arrival time.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

04 Build the solution

A ball is dropped from rest. Complete its downward speeds at 1, 2, 3 s. What is its acceleration?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

At the highest point of a tossed ball, a classmate says gravity stops because speed is zero. Correct the claim.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A fixed-mass cart has acceleration 4 m/s² under 8 N net force.
Find acceleration when net force becomes 12 N.
Explain.

Word help: net force = the vector sum of all forces on one object

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

07 Exit check

A dropped ball falls ideally for 4 s. What is its speed and direction?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Draw three equal-time snapshots of a dropped ball. Then draw a heavier ball released with the same conditions in a vacuum.
Compare the patterns.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3